How Much Targeted Training Does It Take to Top a Multi-Task Leaderboard? Minimum Investment Under Different Aggregation Rules

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Abstract

Public leaderboards rank models by aggregating scores on several sub-datasets, and a developer can try to raise a chosen model's rank by training on data related to some of them. How much such targeted investment is needed to reach first place, and how does it depend on the aggregation rule? We measure investment as the score points added on chosen sub-datasets up to a cap, by the number of sub-datasets touched (K), the total points (L_1) and the largest single jump (L_∞), and compute the minimum exactly for ten rules (mean, power and trimmed means, median, minimum, Borda, plurality, Condorcet winner), with closed forms for the mean, minimum and median. On the Open LLM Leaderboard (4496 models, 6 tasks) and HELM MMLU v1.13.0 (79 models, 57 subjects) we find that (i) with no cap beyond the scale maximum, one sub-dataset with enough headroom suffices under the mean for most targets up to rank 500 on OLLB, but the points needed are n times the gap in mean score; (ii) in points the averaging rules are the most expensive and the median, minimum and plurality rules far cheaper, while ordinal majority rules need the most tasks, so no rule is cheapest on every measure and the order of rules changes with the dataset and the cap; (iii) targeting saves points over a uniform improvement mainly for rules other than the mean (cost 3–51% of the uniform cost for median, minimum and plurality, against 93–100% for the mean at the scale maximum); (iv) the answer depends on the assumed cap: under a cap taken from observed fine-tune gains on one leaderboard, 7.5% of OLLB targets can become first under the mean, and 1.4–12% for the mean across the cap definitions we tried. Investment is measured in score points, not in data or compute, and the conclusions are conditional on the cap, a single deviating developer and the two leaderboards; ordinal rules are solved on a stratified sample.

1 Introduction

Multi-task benchmarks are the main infrastructure for comparing machine-learning models: a leaderboard is a table of scores on several sub-datasets, collapsed into one ranking by an aggregation rule, usually an average. Rankings steer adoption, publication and funding, so developers have a reason to act strategically. One cheap action is to include data related to some sub-datasets in the training of a chosen model, which raises its scores there [Zhou et al., 2023, Yang et al., 2023]. How much of this does it take to reach first place, and does the answer depend on the aggregation rule?

We study this on public leaderboards. We formalise a *targeted investment* (a target model raises its score on chosen sub-datasets up to a cap, all other models fixed), measure it by the number of sub-datasets touched (K), the total score points (L_1) and the largest single jump (L_∞), and compute, *exactly*, the cheapest investment that makes a given model strictly first under each of a menu of rules:

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

mean-type rules, order-statistic rules (trimmed mean, median, minimum) and ordinal rules (Borda, plurality, Condorcet winner). Because the cap is an assumption rather than a measurement, we repeat everything under several cap families, one of them calibrated on observed fine-tune gains.

Contributions. (1) A formal model of targeted investment with three cost measures and an explicit cap assumption, and exact minimum-investment solvers for ten aggregation rules, verified against brute force; closed forms for the mean, minimum and (odd-n) median. (2) An empirical study on two public leaderboards (Open LLM Leaderboard, $M = 4496$ models, $n = 6$; HELM MMLU v1.13.0, $M = 79$, $n = 57$) with intervals over targets, sensitivity to caps, tie margin, evaluation noise, task resampling and data slices. (3) Findings (Section 4.2) that are not forced by the definitions: the order of rules by investment depends on the cost measure, the dataset and the cap, and which rule is easiest to topple under a binding cap flips with the cap; points are cheapest for the median, minimum and plurality, and expensive for averaging rules; targeting saves points mainly for non-mean rules, and for the mean only under binding caps. (4) Code, checksums and a script-to-result map (Appendix A).

What this paper does not do. We do not train models, model rivals' reactions, or claim that any real developer behaved in this way; we also do not solve every rule on every target (ordinal rules use a sample) or compute Copeland, Schulze or Kemeny rankings.

2 Related Work

Multi-task benchmarks and their aggregation. Multi-subject and multi-metric suites such as MMLU [Hendrycks et al., 2020], BIG-Bench Hard [Suzgun et al., 2022] and HELM [Liang et al., 2022] report one score per sub-dataset, and public leaderboards then collapse these scores into a single ranking, most often with an (optionally normalised) arithmetic mean. Dehghani et al. [2021] show that task selection and other benchmark design choices can change which method looks best; our question is complementary: for a *fixed* task set, how robust is the first place to a developer's *action*, and how does that depend on the aggregation rule?

Aggregation as social choice. Ranking models from several score columns is a voting problem, with tasks as voters and models as candidates [Brandt et al., 2016]. Lanctot et al. [2023] advocate social-choice rules for evaluating agents over many tasks. Rofin et al. [2023] apply such rules to re-rank NLP benchmark results and compare the rankings; they do not ask what a model must change to be first. We use the lens to pick rules (Borda, Condorcet-winner requirement, plurality) alongside score-based rules (mean, power means, trimmed mean, median, min), and we ask a question social choice calls *constructive manipulation* or bribery: the cheapest change of one candidate's profile that makes it win [Bartholdi et al., 1989, Faliszewski and Rothe, 2016], or bribery [Faliszewski et al., 2009]. That literature mostly studies the computational complexity of changing ballots; here the changed object is the target's own real-valued scores, within bounds, the score rules reduce to covering problems with closed forms (Propositions 1–3), and the content is how the cost behaves on real score matrices. The Open LLM Leaderboard documents a per-task normalisation of raw scores before averaging [Fourrier et al., 2024]; we take its normalised scores and mean as the leaderboard's own rule.

Gaming, leakage and contamination. Singh et al. [2025] document distortions of the Chatbot Arena leaderboard (private testing of many variants with selective disclosure, unequal data access). Zhou et al. [2023] show that benchmark leakage into training can dramatically boost evaluation results, and Yang et al. [2023] show that rephrased or translated test samples evade n-gram decontamination yet let a model overfit a benchmark. These works establish *that* targeted training on benchmark-related data inflates scores. We take that effect as an assumption (a developer may raise scores on chosen sub-datasets, up to a stated cap) and ask how much of it a given aggregation rule requires to reach first place. Chen et al. [2026] model benchmarking as a Stackelberg game between a designer and developers who allocate post-training resources to benchmark-specific gains, and prove equilibrium results. Their central quantity, the "just-overtake" effort, is the least extra training a model needs to pass the model ranked directly above it on one benchmark; they estimate it from fitted training trajectories in a case study. Their model predicts how much post-training effort developers spend in equilibrium on one benchmark; it has no notion of which sub-dataset to touch or of the aggregation rule. We study *multi-task* leaderboards, ask what it costs to reach *first* place, and compare *aggregation*

rules on real public score matrices, with the cost measured in sub-datasets touched and score points rather than training steps; other models are held fixed and no equilibrium is computed.

3 Setting: Targeted Investment and Aggregation Rules

Leaderboard, rule, target. A leaderboard is a score matrix $X \in \mathbb{R}^{M\times n}$: x_{jt} is the score (in points, 0–100) of model j on sub-dataset t. An aggregation rule R maps X to a ranking. We fix a *target* model i and write $x = x_{i\cdot}$.

Targeted investment. Training on data related to sub-dataset t lets the developer raise the target's score on t to any $y_t \in [x_t, c_t]$, where the *cap* $c_t \geq x_t$ is a modelling assumption (below); scores on other tasks are unchanged by the action, and all other models are fixed (a single deviator; no reaction by rivals). The target is *first* if it is strictly above every rival under R; for score-valued rules f this means $f(y) \geq \tau_i + \varepsilon$ with $\tau_i = \max_{j\neq i} f(x_j)$ and a tie margin $\varepsilon = 10^{-3}$ points, so ties never count as first place. For a rule with no valid solution inside the caps, the target is *infeasible*; infeasibility is reported as a result, never dropped or replaced by a large number.

Measuring investment. For a feasible y let $\delta_t = y_t - x_t$. We report three costs, each minimised separately over feasible y (so the minimisers can differ): the number of tasks touched $K = |\{t : \delta_t > 0\}|$, the total score points $L_1 = \sum_t \delta_t$, and the largest single-task jump $L_\infty = \max_t \delta_t$. K and L_1 are the headline measures; L_∞ gives a *uniform* reference: raising every task by the smallest feasible largest jump L_∞ would cost nL_∞ points, which we compare with the targeted $L_1 \leq nL_\infty$. This reference ignores the cap on individual tasks (a raise of L_∞ may exceed a task's cap or headroom), so it is not itself an allowed investment. "Investment" here is score points under a cap, not training data or compute; we make no claim about how many examples it takes to gain a point.

Caps. The cap is the main assumption and the main sensitivity. With $x_{\max,t} = \max_{j\neq i} x_{jt}$ and all caps clipped at 100: `ceil`: $c_t = 100$ (unconstrained reference); `front`h: $c_t = \max(x_t, x_{\max,t} + h)$ (the target may at most overtake the best rival on t by h points, $h \in \{0, 2\}$); `gain`h: $c_t = x_t + h$ (at most h points per task, $h \in \{5, 10, 20\}$); `emp`q: $c_t = x_t + g_t^{(q)}$ where $g_t^{(q)}$ is the q-th percentile ($q \in \{95, 99\}$) of observed fine-tune-minus-base score differences on task t, measured on 540 base/fine-tune pairs of the Open LLM Leaderboard (Section 4).

Rules. *Score rules* (rank by $f(x_j)$): the arithmetic mean; the geometric mean and power means $\left(\frac{1}{n}\sum_t (x_t+1)^p\right)^{1/p} - 1$ with $p < 1$, which interpolate between the mean ($p = 1$) and the minimum ($p \to -\infty$); the trimmed mean (drop the k smallest and largest scores); the median; the minimum. *Ordinal rules* (use only within-task orderings): Borda (points per task = number of models strictly below plus half the ties, summed over tasks); plurality (number of tasks on which the model is first, ties split); and the *Condorcet-winner requirement*, i.e. the target must beat every rival on more than half of the tasks. Being the Condorcet winner is *sufficient* for first place under any Condorcet-consistent rule (Copeland, Schulze, Kemeny) but not necessary, so its minimum investment is an upper bound for those rules [Brandt et al., 2016]. On the Open LLM Leaderboard the deployed rule is the mean of normalised task scores; this is the headline rule there.

Exact solvers. For each rule we compute exact minima: closed forms or greedy/water-filling for the mean and power means, order-statistic arguments for the median and minimum, and mixed-integer programs (HiGHS) for the trimmed mean and the ordinal rules; every solver is checked against brute-force enumeration on small random instances (code and tests in the supplement).

3.1 Closed forms

Write n tasks, headrooms $h_t = c_t - x_t$, and let τ and ε be as above.

Proposition 1 (Mean)**.** *Under the arithmetic mean let $N = n(\tau + \varepsilon) - \sum_t x_t$ (the points needed). If $N \leq 0$ the target is already first. Otherwise the target can become first iff $\sum_t h_t \geq N$; then $\min L_1 = N = n$ (gap in mean score $+ \varepsilon$) independently of how the points are spread, $\min K$ is the smallest k such that the k largest headrooms sum to at least N, and $\min L_\infty$ is the λ solving $\sum_t \min(h_t, \lambda) = N$.*

Proof. $\text{mean}(y) \geq \tau + \varepsilon$ iff $\sum_t \delta_t \geq N$ with $0 \leq \delta_t \leq h_t$. This is a covering constraint: the minimum of $\sum \delta_t$ is N if feasible; the fewest nonzero δ_t is attained by filling the largest headrooms first; and $\max_t \delta_t$ is minimised by $\delta_t = \min(h_t, \lambda)$. □

Proposition 2 (Minimum)**.** *Under the minimum rule let $b = \tau + \varepsilon$. The target can become first iff $c_t \geq b$ for every t with $x_t < b$; then* $\min K = |\{t : x_t < b\}|$ *and* $\min L_1 = \sum_t (b - x_t)^+$.

Proof. $\min_t y_t \geq b$ iff $y_t \geq b$ for all t; tasks with $x_t \geq b$ need no change and every other task must be raised to at least b. □

Proposition 3 (Median, odd n)**.** *Under the median with n odd let $b = \tau + \varepsilon$ and $m = (n+1)/2$. The target is first iff at least m tasks end at or above b; hence* $\min K = \max(0, m - |\{t : x_t \geq b\}|)$ *if enough tasks with $x_t < b$ have $c_t \geq b$ (else infeasible), and* $\min L_1$ *is the sum of the* $\max(0, m - |\{t : x_t \geq b\}|)$ *smallest shortfalls $b - x_t$ among them.*

Proof. For odd n, $\text{median}(y) \geq b$ iff at least m entries are $\geq b$. Tasks already above b count for free; each further task costs at least its shortfall, and raising exactly to b suffices. □

For even n (the Open LLM Leaderboard has $n = 6$) the median is the mean of two middle order statistics, and for the other rules no simple closed form exists; the solvers handle them, and Section 4.2 checks the closed forms above against the solver output on every target. Propositions 1–3 already show where the rules differ: the mean pays for *points* (n times the gap) and can place them on a single task (K can be 1) or spread them evenly at the same cost in points, whereas the minimum and the median pay for *specific entries*: K is the number of tasks on which the target is below the bar b (minimum), respectively m minus the number already above it (median), while the points needed can be small. One might therefore expect breadth-demanding rules (median, minimum) to need the most investment, the minimum to need all n tasks, and targeting to pay off most for the mean. Proposition 2 shows that the second expectation is false; Section 4.2 shows from the data that the first and third fail as well.

4 Experiments

4.1 Data, targets and uncertainty

We use two public score sources, both frozen snapshots; checksums and the full provenance are in Appendix A. (i) The **Open LLM Leaderboard** dataset `open-llm-leaderboard/contents` (commit `9c09a7ca`, last updated 2025-03-20): six tasks (IFEval, BBH, MATH Lvl 5, GPQA, MuSR, MMLU-PRO), scores in the leaderboard's *normalised* form, whose mean is the leaderboard's own rule; after removing duplicate model names and one flagged submission, $M = 4496$ models. (ii) The **HELM** MMLU release `v1.13.0` (2025-01-10): exact-match scores on the $n = 57$ MMLU subjects for the $M = 79$ models that have all of them. Every model is a target, and we report results by the target's rank under the plain mean (groups 2, 3–5, 6–10, 11–30, 31–100, 101–500, >500; HELM has no model beyond rank 79). A group's number is the median over its targets, with a 95% percentile-bootstrap interval over targets ($B = 1000$, seed 0); a target that is infeasible under a cap counts as $+\infty$ in that median and we give the infeasible share, and "over feasible" medians are labelled as such. Costs of 0 mean the target is already first under that rule. Score rules are solved for all targets and all caps. Ordinal rules are expensive (mixed-integer programs) and are solved on a rank-stratified sample of 136 OLLB targets (ranks 2–21 plus 150 log-uniform draws in rank, seed 0) and all 78 non-leading HELM models; population statements about the sample use stratum weights and are flagged. Exact Copeland, Schulze and Kemeny rankings were not computed.

Caps. We show `ceil` (reference upper bound on what is possible), `emp95` and `gain10`. The empirical cap `emp95` is the 95th percentile of the fine-tune-minus-base score difference per OLLB task, measured on 540 base/fine-tune pairs (154 labelled chat models, 346 fine-tuned on domain-specific datasets, 40 other); it exists only for OLLB. Its values in points are 46.0 (IFEval), 7.3 (BBH), 21.3 (MATH Lvl 5), 3.8 (GPQA), 7.2 (MuSR) and 3.2 (MMLU-PRO), the same for every target; the large IFEval and MATH values may reflect general instruction tuning rather than training on related data. Without the 154 chat-typed pairs they fall to 44.6 and 19.4 and without chat-named pairs as well they are 55.7 and 18.8 (Appendix C); the self-declared type does not identify instruction conversion,

Table 1: Median minimum L_1 (total points) [95% bootstrap interval] over the *feasible* targets of each rank group, per rule and cap; "a/b feas." means that a of the group's b targets can become first under the cap (cells without it: all can), "all inf." that none can; †: median over fewer than five feasible targets, its interval is unreliable. The targets are all models of the leaderboard (a census), so the interval shows how much the median depends on which models are in the group, not sampling from a larger population; medians of different caps are over different feasible subsets. The group >500 is 89% of the targets; under the binding caps it is almost wholly infeasible. Ordinal rules (*) use the stratified sample (smaller b). Open LLM Leaderboard; the corresponding table for K and for HELM MMLU are in Appendix B.

rule	rank 6-10	rank 31-100	rank 101-500	rank >500
cap: `ceil`				
	(n=5)	(n=70)	(n=400)	(n=3996)
mean	28 [25, 28]	53 [51, 54]	75 [73, 77]	188 [186, 190]
geometric mean	22 [18, 24]	37 [33, 38]	51 [50, 53]	160 [158, 162]
trimmed mean	22 [21, 24]	44 [43, 45]	55 [54, 56]	146 [145, 148]
median	0.3 [0.0, 5.0]	18 [16, 21]	29 [28, 30]	111 [110, 113]
min	13 [6.2, 18]	14 [11, 17]	12 [11, 13]	47 [46, 47]
Borda*	5.8 [5.0, 8.9]	16 [14, 20]	34 [31, 39]	122 [100, 142]
Condorcet winner*	12 [10, 16]	26 [24, 30]	45 [42, 50]	111 [99, 123]
plurality*	20 [15, 23]	23 [21, 25]	29 [27, 33]	45 [43, 49]
cap: `emp95`				
	(n=5)	(n=70)	(n=400)	(n=3996)
mean	28 [25, 28]	53 [51, 54]	71 [69, 72] (236/400 feas.)	all inf. (3996)
geometric mean	41 [25, 50]	48 [42, 50] (28/70 feas.)	73 [70, 77] (60/400 feas.)	all inf. (3996)
trimmed mean	22 [21, 24]	34 [30, 36] (18/70 feas.)	58 [47, 64] (17/400 feas.)	all inf. (3996)
median	0.3 [0.0, 5.0]	20 [14, 22] (37/70 feas.)	21 [19, 24] (73/400 feas.)	31 [27, 49]† (4/3996 feas.)
min	6.2 [6.2, 6.2]† (1/5 feas.)	3.9 [1.8, 5.8] (14/70 feas.)	5.8 [3.8, 7.0] (25/400 feas.)	20 [10, 21] (15/3996 feas.)
Borda*	5.8 [5.1, 9.2]	16 [13, 21] (28/35 feas.)	33 [30, 38] (18/32 feas.)	all inf. (40)
Condorcet winner*	12 [12, 12]† (1/5 feas.)	25 [22, 29] (5/35 feas.)	all inf. (32)	all inf. (40)
plurality*	19 [15, 23]† (4/5 feas.)	27 [24, 31] (16/35 feas.)	27 [25, 34] (6/32 feas.)	all inf. (40)
cap: `gain10`				
	(n=5)	(n=70)	(n=400)	(n=3996)
mean	28 [25, 28]	53 [51, 54]	59 [59, 59] (14/400 feas.)	all inf. (3996)
geometric mean	24 [18, 29]	42 [38, 49] (53/70 feas.)	55 [52, 56] (60/400 feas.)	all inf. (3996)
trimmed mean	22 [21, 24]	33 [31, 36] (20/70 feas.)	39 [35, 42] (6/400 feas.)	all inf. (3996)
median	0.3 [0.0, 5.0]	15 [15, 17] (46/70 feas.)	23 [21, 23] (125/400 feas.)	23 [22, 32]† (3/3996 feas.)
min	13 [6.2, 18]† (4/5 feas.)	8.7 [6.9, 12] (46/70 feas.)	9.7 [9.4, 10] (280/400 feas.)	14 [13, 16] (98/3996 feas.)
Borda*	5.8 [5.0, 8.9]	17 [13, 20] (33/35 feas.)	32 [29, 35] (20/32 feas.)	all inf. (40)
Condorcet winner*	12 [10, 16]	24 [22, 27] (18/35 feas.)	37 [37, 37]† (1/32 feas.)	all inf. (40)
plurality*	all inf. (5)	16 [14, 18]† (2/35 feas.)	all inf. (32)	all inf. (40)

so `emp95` is a cap on the large gains of fine-tunes and not a measurement of targeted training. `front2` (overtake the best rival on a task by at most 2 points) and the remaining caps are in the appendix.

4.2 Results

Closed forms and solvers agree. For the mean, minimum (both datasets) and median (HELM; n odd) the closed forms of Propositions 1–3, coded independently of the solvers, agree with the solver output on every instance (largest absolute difference in L_1 2.3×10^{-13}; feasibility and K agree in 100% of 35,960 OLLB and 468 HELM mean instances, and in the other checked cells), and on the 95,181 OLLB and 2,967 HELM instances of all rules and caps $L_\infty \le L_1 \le nL_\infty$ and $1 \le K \le n$ hold. For these three rules the solver is itself a closed-form or greedy routine, so the comparison shows that two implementations agree; the solvers, including the mixed-integer ones, were in addition checked against brute-force enumeration on small random instances. The minimum rule does *not* need all tasks (Proposition 2): it needs the tasks below the best rival's minimum ($K = 2$ in the median at ranks 6–500 on OLLB; 2, 7 and 17 at ranks 6–10, 11–30, 31–100 on HELM).

The rule changes the cost, and the ordering of rules depends on the cost measure and the dataset. Figure 1 and the first numbers below are for the uncapped case. One might expect the rules that need breadth (median, minimum) to need the most investment; in points the opposite holds, in tasks it does not. In total points (L_1), averaging rules are the *most* expensive: at ranks 101–500 on OLLB the median L_1 is 75 points for the mean, 55 for the trimmed mean and 51 for the geometric mean, against 29 for the median rule and 12 for the minimum; at ranks 31–100 on HELM it is 1021 for the mean and 992 for the geometric mean against 207 for the median rule, 200 for the minimum

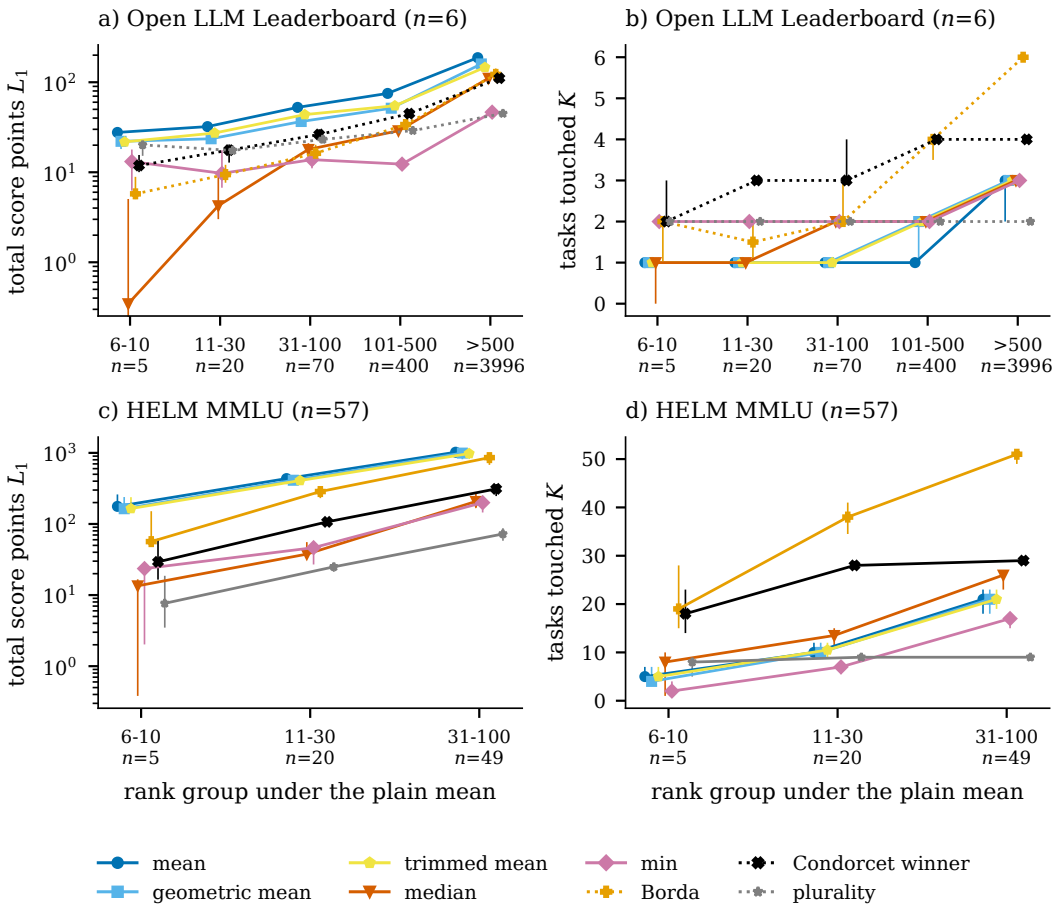


Figure 1: Minimum investment to reach first place against the target's rank group under the plain mean, cap `ceil` (every task may be raised to the scale maximum, so all targets are feasible). Left: total points L_1 (log scale); right: tasks touched K. Top: Open LLM Leaderboard (4496 models, 6 tasks); bottom: HELM MMLU v1.13.0 (79 models, 57 subjects). Points are group medians, bars 95% bootstrap intervals over targets; dotted lines mark the ordinal rules, which on OLLB use the stratified sample of 136 targets. Ranks 2 and 3–5 (one and three targets) are in the tables of the appendix.

and 72 for plurality. The reason is Proposition 1: the mean must make up n times the gap in mean score, while median and minimum only have to lift specific entries to the best rival's median or minimum. In tasks touched (K), the order reverses for the mean: its median K is 1 for every OLLB group up to ranks 101–500 (one sub-dataset with enough headroom suffices), while on HELM at ranks 31–100 the median rule needs 26 subjects (21 for the mean, 17 for the minimum) and the ordinal rules need the most: Borda 51 of 57, the Condorcet requirement 29, plurality 9. No rule is cheapest on every measure. The order of rules by median L_1 is also not fixed. Across caps on the same dataset the agreement is only moderate on both datasets (Figure 4a; Kendall τ between the rule orders of two caps, for rank groups of at least five targets, 6–10 to 31–100: median 0.45, minimum 0.22 on HELM, 40 pairs; median 0.50, minimum -0.52 on OLLB, 77 pairs, and 0.15 at ranks 101–500, 21 pairs), it differs further between the two datasets (median 0.36, range -0.50 to 0.86, 131 pairs), and under evaluation-noise and task-bootstrap resampling the mean rank of a rule moves by one to several places (Figure 9b,c; mean τ to the median ordering 0.74–0.77 on HELM `gain10`, 0.75 under noise and 0.35 under task bootstrap on OLLB). We therefore report the measured orders with their intervals, not a universal order; what is invariant are the closed forms.

Rank, and infeasibility under realistic caps. At `ceil`, the median L_1 of the largest-rank group is the largest for every rule (Figure 1), and the group medians grow with the rank group for the arithmetic, geometric and trimmed mean on OLLB and for all rules on HELM except plurality, whose

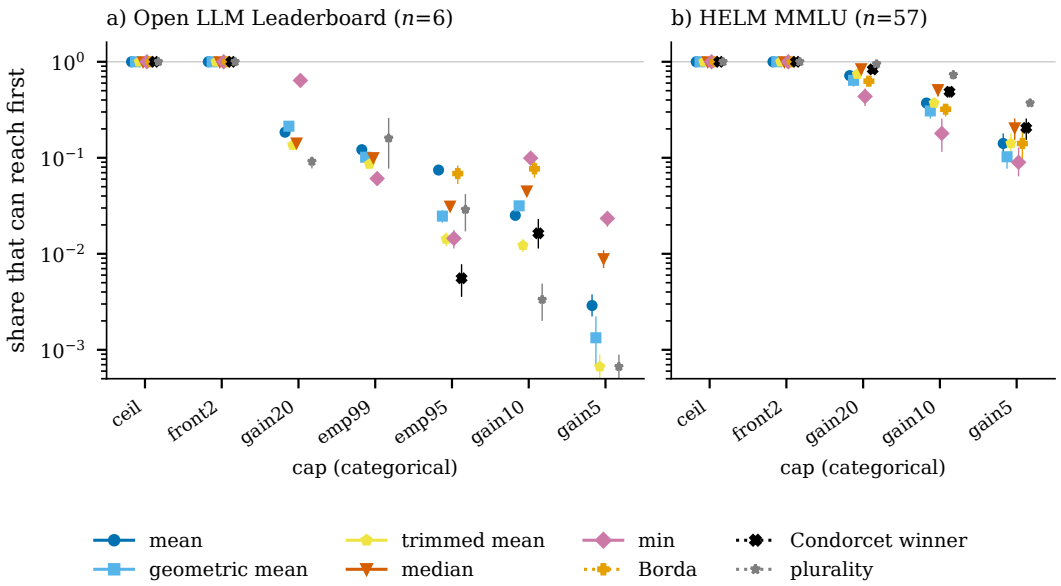


Figure 2: Share of all targets that can reach first place, by cap and rule (bars: percentile bootstrap 95% intervals over targets, within strata for the sampled ordinal rules; zero shares are drawn at the axis floor; caps are categorical, roughly from loosest to tightest). (a) OLLB (4495 targets, 89% of them ranked below 500): 1.4–7.5% at `emp95`, 1.2–9.9% at `gain10`, 14–64% at `gain20`, depending on the rule. (b) HELM MMLU (78 targets): 18–73% at `gain10`, 6–37% at `gain5`. Every target is feasible at `ceil` and `front2`. Ordinal rules (dotted) on OLLB: 136-target stratified sample re-weighted by stratum size. By rank group: Figure 6.

first two groups hold one and three targets. They do not for the median rule, the minimum rule, the harmonic mean and the power mean $p = -5$ on OLLB: for the minimum, the group medians from rank 2 to >500 are 3.7, 3.1, 13.1, 9.7, 13.8, 12.3 and 46.5 points. Under the mean, $L_1 = n \cdot (\text{gap} + \varepsilon)$ (Proposition 1), so its growth with rank is that of the gap to the leader in the leaderboard's score distribution, not an effect of the rule. As a descriptive summary only, the log–log regression of L_1 on rank (feasible targets, `ceil`) has slopes from 0.44 (mean) to 0.65 ($p = -5$) on OLLB and 1.07 (geometric mean) to 2.17 (median) on HELM; a power law has the smallest relative error in 25 of 32 OLLB and 16 of 18 HELM group-level fits, but a linear form has the higher R^2 on the L_1 scale in 14 of 32 and 17 of 18 (HELM: four or five groups; Appendix C), so "roughly a sub-linear power of the rank" on OLLB is a description, not a demonstrated law; the slopes of the two leaderboards differ unexplained. With `emp95` the mean rule is feasible for all targets up to rank 100, 59% at ranks 101–500 and none beyond rank 500 on OLLB, while the minimum rule is feasible for 20% of ranks 6–10 and 6% of ranks 101–500; the share of all OLLB targets that can become first under `emp95` is 7.5% for the mean and 1.4% for the minimum (`gain10`: 2.5% and 9.9%; Figure 2). Which rule is easiest to topple therefore flips with the cap.

Targeted versus uniform investment. Comparing the targeted cost L_1 with the cost nL_∞ of a uniform improvement by the same largest jump gives ratios of 1.00 (OLLB) and 0.93–0.99 (HELM) for the mean. This follows from Proposition 1 (the mean needs n times the gain in mean score wherever the points go) and is not a finding; targeting saves tasks, not points. Under binding caps the mean's ratio falls to group medians of 0.42–0.89 on OLLB with `emp95` and 0.61–0.70 on HELM with `front0` (0.42–1.00 over all caps), but since nL_∞ ignores the cap, this mixes the effect of the cap on L_∞ with that of targeting; we did not compute a cap-respecting uniform raise. For the other rules the ratios are far below one (ranges over rank groups at `ceil`, OLLB and HELM): 0.33–0.51 and 0.09–0.24 for the median, 0.26–0.43 and 0.03–0.13 for the minimum, 0.26–0.29 and 0.09–0.13 for plurality.

Conditions: cap, number of tasks, data slices, ties. The cap is the assumption on which the answer depends most (Figures 2 and 6): a runner-up needs a median of 4.8 points under the mean

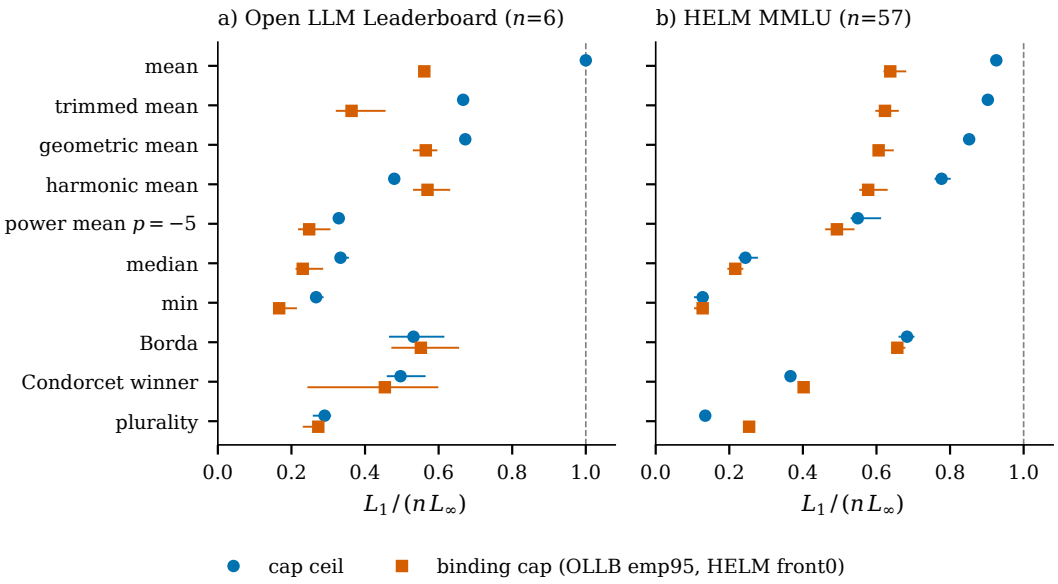


Figure 3: Targeted versus uniform investment: ratio $L_1/(nL_\infty)$ of the cheapest targeted investment to the cost of improving all n tasks by the same largest jump, rank group 31–100 (OLLB $n = 70$, HELM $n = 49$; median over feasible targets, 95% bootstrap interval), at `ceil` and at one binding cap per dataset (OLLB `emp95`, HELM `front0`). A ratio of one (dashed) means that targeting saves no points; $L_1 \leq nL_\infty$ always holds, and nL_∞ ignores the cap. Ordinal rules on OLLB: stratified sample; under `emp95` the ratio is over the feasible targets only (Figure 2).

at `ceil` on OLLB, yet beyond rank 500 no target can become first under the mean with `emp95` or `gain10`, and at ranks 101–500 the shares are 59% and 4% (Figure 6). Without chat-typed pairs the shares under `emp95` are 6.7% (mean), 2.7% (median) and 1.7% (minimum), against 7.5, 3.1 and 1.4%; a cap on the share of the headroom a fine-tune closed gives 1.4, 1.7 and 0.8%, and a flat quarter of the headroom 12.2, 9.4 and 100% (Table 8). The number of tasks matters: on random subsets of n HELM subjects (Figure 4b) the log–log slope of the median L_1 in n is 0.55–1.06 for the mean but -0.14 to 0.06 for median and minimum (five subset sizes, one benchmark family). Removing merged models or keeping the top 500 leaves the rule order close to the main one (Kendall τ 0.81–1.00); raw instead of normalised OLLB scores changes it more (τ 0.71 at `ceil`, 0.52 at `gain10`; Figure 9d). Raising the tie margin from 10^{-3} to 0.1 points moves the median L_1 by 0.6–3.3% in the 20 checked rule–cap cells, and counting ties as losses leaves it unchanged, but individual targets can change by factors up to 6.0 for Borda and 2.4 for the mean (Table 7).

5 Discussion and Limitations

What the results say. On both leaderboards, first place under the deployed averaging rules is cheap in tasks for top-ranked targets and expensive in points: a single sub-dataset can suffice, but the points needed under the mean are n times the gap to the leader and grow steeply with the target's rank (log–log slope 0.4–0.7 on OLLB and 1.1–2.2 on HELM at `ceil`, as a descriptive summary). Whether other rules are easier or harder depends on what one counts. Median, minimum and plurality need far fewer points, and targeting pays for them (cost 3–51% of the uniform improvement), while the mean is indifferent to where the points go; rules that need a majority of pairwise wins (Borda, Condorcet requirement) need many tasks. Realistic caps change the picture more than the rule does: under `emp95` or `gain10`, at most 2.5% of the OLLB targets beyond rank 500 can win under any score rule (none under the mean), but `gain20` lets 60% of them win under the minimum rule. The results suggest that a designer who wants a leaderboard to resist targeted investment would do better to bound what can be gained per sub-dataset (for example by holding data out) than to rely on the choice of aggregation rule alone; we did not test such a defence beyond the clipping of Appendix C, which leaves the cost of first place unchanged, and we do not claim that any rule is robust.

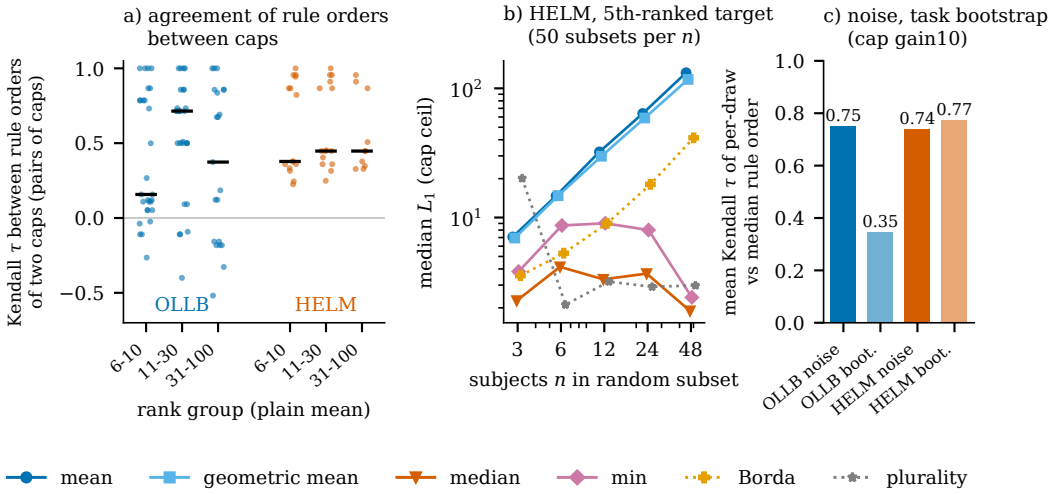


Figure 4: Instability of the rule order and robustness. (a) Kendall τ between the rule orders (by median L_1 in the rank group; at least seven common rules) of every two caps of the same dataset: one dot per pair of caps (OLLB 28 pairs, 21 for ranks 31–100; HELM 15, 10), bar: median. (b) HELM, median L_1 at `ceil` against the number n of subjects in a random subset (50 draws per n), target ranked fifth in the subset. (c) Mean Kendall τ of the per-draw rule order with the order of medians under evaluation noise and under task bootstrap (100 draws; `gain10`; the ten best models after the leader as targets). Intervals and per-rule ranks: Figure 9.

Limitations. (i) *The cap is an assumption.* The empirical cap uses 540 fine-tune/base pairs of one leaderboard, 154 labelled chat models and 346 fine-tuned on domain-specific datasets, so it mixes generic instruction tuning with the targeted training we want to model (dropping chat-typed pairs changes the feasible shares by under a point, but they range from 0.4 to 16% with the rule and the definition, and the minimum rule reaches 100% under a flat quarter of the headroom; Appendix C); it is the same for every target whatever its starting score; and it says nothing about what a determined developer with new data could reach (the leakage studies we cite report larger gains), so it may understate as well as overstate. "Investment" is score points, not data or compute. (ii) *Single deviator.* Other models do not react, and several developers acting together or a designer changing the rule is outside the model. (iii) *Small n on OLLB.* With six tasks, rank rules are coarse, and the even-n median has no simple closed form; the HELM results (one benchmark family, 79 models, 57 correlated subjects with 100 to 1534 test items each, so one item is 1.0 point at the smallest) should not be generalised to other leaderboards, and K counts six distinct benchmarks on OLLB but 57 subjects of one benchmark on HELM, so we compare K within a dataset, not across. (iv) *Sampled ordinal rules.* Borda, Condorcet requirement and plurality on OLLB were solved for a stratified sample of 136 targets, not all 4496, and no exact Copeland, Schulze or Kemeny ranking was computed; the Condorcet requirement is only an upper bound for those rules. (v) *Stability.* Rule orders differ between the two leaderboards and, on OLLB with six tasks, under task resampling, and agree only moderately across caps (Kendall τ about 0.5; Section 4.2); the τ values have no null distribution (not computed), and individual targets can change cost substantially with the tie margin. (vi) *Scores are frozen snapshots*, the data carry no declared licence, and the models' scores were taken as given, including possible contamination. (vii) *Compute:* 7.7 CPU-hours in total, above the task's rough 4-hour estimate (Appendix A).

Broader impact. The analysis describes how cheaply rankings can be moved by targeted training, which could inform gaming; it uses only public scores and trains no model, and its practical content is for benchmark designers (cap what a sub-dataset can contribute, report intervals). We make no claim about any real developer.

References

John J. Bartholdi, Craig A. Tovey, and Michael A. Trick. The computational difficulty of manipulating an election. *Social Choice and Welfare*, 6(3):227–241, 1989.

Felix Brandt, Vincent Conitzer, Ulle Endriss, Jérôme Lang, and Ariel D. Procaccia, editors. *Handbook of Computational Social Choice*. Cambridge University Press, 2016. doi: 10.1017/CBO9781107446984.

Yatong Chen, Guanhua Zhang, and Moritz Hardt. Leaderboard incentives: Model rankings under strategic post-training. *arXiv preprint arXiv:2603.08371*, 2026.

Mostafa Dehghani, Yi Tay, Alexey A. Gritsenko, Zhe Zhao, Neil Houlsby, Fernando Diaz, Donald Metzler, and Oriol Vinyals. The benchmark lottery. *arXiv preprint arXiv:2107.07002*, 2021.

Piotr Faliszewski and Jörg Rothe. Control and bribery in voting. In Felix Brandt, Vincent Conitzer, Ulle Endriss, Jérôme Lang, and Ariel D. Procaccia, editors, *Handbook of Computational Social Choice*, pages 146–168. Cambridge University Press, 2016.

Piotr Faliszewski, Edith Hemaspaandra, and Lane A. Hemaspaandra. How hard is bribery in elections? *Journal of Artificial Intelligence Research*, 35:485–532, 2009. doi: 10.1613/jair.2676.

Clémentine Fourrier, Nathan Habib, Alina Lozovskaya, Konrad Szafer, and Thomas Wolf. Open LLM leaderboard v2. Hugging Face, `https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard`, 2024.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2020.

Marc Lanctot, Kate Larson, Yoram Bachrach, Luke Marris, Zun Li, Avishkar Bhoopchand, Thomas Anthony, Brian Tanner, and Anna Koop. Evaluating agents using social choice theory. *arXiv preprint arXiv:2312.03121*, 2023.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, Benjamin Newman, Binhang Yuan, Bobby Yan, Ce Zhang, Christian Cosgrove, Christopher D. Manning, Christopher Ré, Diana Acosta-Navas, Drew A. Hudson, Eric Zelikman, Esin Durmus, Faisal Ladhak, Frieda Rong, Hongyu Ren, Huaxiu Yao, Jue Wang, Keshav Santhanam, Laurel Orr, Lucia Zheng, Mert Yuksekgonul, Mirac Suzgun, Nathan Kim, Neel Guha, Niladri Chatterji, Omar Khattab, Peter Henderson, Qian Huang, Ryan Chi, Sang Michael Xie, Shibani Santurkar, Surya Ganguli, Tatsunori Hashimoto, Thomas Icard, Tianyi Zhang, Vishrav Chaudhary, William Wang, Xuechen Li, Yifan Mai, Yuhui Zhang, and Yuta Koreeda. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*, 2022.

Mark Rofin, Vladislav Mikhailov, Mikhail Florinsky, Andrey Kravchenko, Tatiana Shavrina, Elena Tutubalina, Daniel Karabekyan, and Ekaterina Artemova. Vote'n'rank: Revision of benchmarking with social choice theory. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, 2023. doi: 10.18653/v1/2023.eacl-main.48.

Shivalika Singh, Yiyang Nan, Alex Wang, Daniel D'Souza, Sayash Kapoor, Ahmet Üstün, Sanmi Koyejo, Yuntian Deng, Shayne Longpre, Noah A. Smith, Beyza Ermis, Marzieh Fadaee, and Sara Hooker. The leaderboard illusion. *arXiv preprint arXiv:2504.20879*, 2025.

Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi, Denny Zhou, and Jason Wei. Challenging BIG-Bench tasks and whether chain-of-thought can solve them. *arXiv preprint arXiv:2210.09261*, 2022.

Shuo Yang, Wei-Lin Chiang, Lianmin Zheng, Joseph E. Gonzalez, and Ion Stoica. Rethinking benchmark and contamination for language models with rephrased samples. *arXiv preprint arXiv:2311.04850*, 2023.

Kun Zhou, Yutao Zhu, Zhipeng Chen, Wentong Chen, Wayne Xin Zhao, Xu Chen, Yankai Lin, Ji-Rong Wen, and Jiawei Han. Don't make your LLM an evaluation benchmark cheater. *arXiv preprint arXiv:2311.01964*, 2023.

A Reproducibility and Provenance

Everything in the paper is produced by the code in the delivery (directory `code/`, with `code/README.md` as the full reproduction notes, and `figtab/` for figures and tables). Every number quoted in the text comes from the compact result files in `code/results/analysis/`, and every figure and table of the paper is rebuilt from them (`figtab/scripts/make_all.sh`). The per-instance solver results (one row per target, cap and rule; about 209 MB, 12 MB gzip-compressed) are in the delivery as `code/results/instances/*.csv.gz`, except the one large file `ollb_powmeans_weighted.csv` (131 MB), which `reproduce.sh` regenerates. A manifest (`code/results/MANIFEST.md`) lists the SHA-256 and a one-line convention of every result file. `bash code/reproduce.sh` (offline, CPU only) unpacks the instance files, regenerates the large file, rebuilds all of `code/results/analysis/`, runs the closed-form checks and compares every file with the manifest; on a 2-CPU container it took 1844 s wall (1575 s CPU), and 89 of 91 files were byte-identical, the other two differing only in a measured CPU-time column that the manifest hash excludes. This path rebuilds the analysis tables and checks from the per-instance results; it does not re-run the ordinal mixed-integer, HELM, noise and bootstrap solver jobs that produced them, which `make experiments` does. `paper/scripts/check_numbers.py` checks that the numbers quoted in the text agree with the compact result files.

Data. Table 2 gives the versions and full checksums of the inputs. The processed score matrices (79 and 4497 rows) are committed with their checksums and were rebuilt byte-identically from the raw files. The Open LLM Leaderboard `contents` file has 4576 rows; we drop 79 rows whose model name (`fullname`) is duplicated, keeping the best-scoring variant, and one flagged submission, which leaves 4496 models; all 79 are float16/bfloat16 pairs of one model, and keeping the first or the latest submission, or the worst-scoring one, leaves the leader and runner-up unchanged and the group medians of the headline cells within 5.3% (Appendix C). The HELM release is a versioned snapshot; the download date was not recorded beyond November 2026, and the manifest hash identifies the content. The evaluation-noise model on OLLB needs only per-task item counts, which we took from one results file because every model is evaluated on the same items of each task (one count vector in 60 randomly sampled models; the file is pinned in the table).

Table 2: Inputs, versions and SHA-256 checksums.

Input	Version	SHA-256
OLLB `contents`	dataset `open-llm-leaderboard/contents`, file `data/train-00000-of-00001.parquet`, commit `9c09a7cae43334062a82cb164f2ef255013dafa2`, last updated 2025-03-20	`f13a7061e63133bc8745d89f8c7a56148584940ede22d3b27e21cad1e078dca5`
HELM MMLU	release v1.13.0 (2025-01-10), 303 files; per-file hashes in `code/data/processed/`	manifest file: `830bdb10a5518c50689d60c86623cc3d79ff5317d7e6b96408ea7dc2058de12b`
OLLB item counts for the noise model	one `results` file, `0-hero/Matter-0.2-7B-DPO`, dataset commit `aa81ecc38fdc5708254b833923368970efdf5ef5`; built into `ollb_noise.json`	`4e0b5eb02ed53911c54d9044a47e3fd901d244349237372cbfbacd3c0da721a4`

Environment, seeds and commands. Python 3.11.2, NumPy 2.4.6, SciPy 1.17.1 (bundled HiGHS), pandas 3.0.6, pyarrow 26.0.0; CPU only. All runs are deterministic given the seeds: rank-stratified target sample seed 0 (ranks 2–21 plus 150 log-uniform in rank, 136 distinct targets), Dirichlet weights seed $1000 + k$, Monte Carlo draws seed 0 (100 draws for evaluation noise and task bootstrap, 50 for task subsets), bootstrap over targets $B = 1000$ seed 0. The commands are `make tests`, `make experiments`, `make analysis`, `make theory` and `make manifest` (see `code/README.md`); the ten solver unit tests compare the solvers with brute force on small random instances.

Compute. The measured solver CPU time, summed over 260,572 solves, is 3.25 CPU-hours (OLLB score rules on all 4496 targets and eight caps 0.26; OLLB power and weighted means 0.06;

OLLB Borda and Condorcet on the 136-target sample 1.78; plurality 0.13; HELM 0.27; slices, ε and tie variants 0.77). There were no solver time-outs, so no censoring correction was needed. The scheduler's accounting of all jobs of the study, including data builds, tests, aborted and repeated runs, is about 7.7 CPU-hours, above the rough 4 CPU-hour estimate in the task statement; the required grid fits in the estimate, the robustness extras and development reruns do not. The job logs sum to about 4.9 hours of wall time on a shared, oversubscribed host, so wall time and CPU time are not comparable; re-running all jobs with `make experiments` is a matter of hours (the scheduler's total is 7.7 CPU-hours), not days.

Checks of the closed forms. The closed forms of Propositions 1–3 were coded independently of the solvers and compared with solver output on every instance (`code/results/theory_checks.txt`). Feasibility and K agree on all instances of the mean, minimum and median (HELM) and the mean and minimum (OLLB); the largest absolute difference in L_1 is 2.3×10^{-13}. The bounds $L_\infty \le L_1 \le nL_\infty$ and $1 \le K \le n$ hold on all 98,148 instances, and the identity $L_1 = n(\text{gap} + \varepsilon)$ for the mean holds to machine precision over 499 random task subsets.

Provenance. All code, experiments, analyses and this manuscript were produced by AI agents (Claude models) working autonomously in separate roles (context, planning, analysis, presentation, writing); no human wrote or ran any of it. The study plan from which the questions of Section 4.2 come was written after pilot runs existed and is not a pre-registration.

Script-to-output map. Table 3 lists which script produces each figure and table (figure and table numbers are those of the PDF); the analysis scripts read the result files named in `code/README.md` (section 6).

Table 3: Producing script of each figure and table.

Output	Script	Inputs
Figures 1–4, 5–10	`figtab/scripts/fig*.py`, `main_*.py` (all: `make_all.sh`)	`code/results/analysis/*.csv`, `theory_pairs.csv`; hashes in `figtab/INPUTS.md`
Tables 1, 4–6, 7	`figtab/scripts/tables.py`, `tab_eps_ties.py`	`summary_groups.csv`, `robust_eps_ties.csv` in `code/results/analysis/`
Headline numbers of the text	`code/src/analyze.py`, `analyze_robust.py`	`code/results/analysis/*.csv` (checked by `paper/scripts/check_numbers.py`)

Licences. The data are public leaderboard scores. The HELM storage bucket and the OLLB `contents` and `results` datasets declare no dataset licence (the HELM code is Apache-2.0; the OLLB `contents` card has a per-model licence column only). We used only aggregate scores for research analysis and did not redistribute raw files; the derived score matrices committed with the code carry their sources and checksums. We treat this as permissible research use of public data; this is our judgement, not a legal opinion, and anyone redistributing the processed matrices should check the upstream terms.

Statement on withheld material. The task statement names a published paper on this question that must not be searched for, accessed or used. We did not search for it, and no one involved reported having met it. The analysis made no literature or web search (only the scripted data downloads from the two hosts above). Literature searches for the related work (social-choice bribery, manipulation and control, leaderboard sensitivity, benchmark aggregation) returned no item identified as that paper, and no hit was opened beyond a title or abstract snippet except the full text of the starter-list item of Chen et al. [2026]; for a work on the sensitivity of language-model leaderboards that a search returned we saw only its title, authors and an abstract snippet, did not read it, and do not cite it. We cannot rule out that a page we did not read is related to the withheld paper.

B Additional Tables and Figures

Tables 4–6 complete Table 1; Table 7 gives the sensitivity to the strictness margin and to ties; Figures 5–10 show the cost and feasibility under the empirical cap, feasibility by rank group, the cost under different caps, the power-mean continuum, the full robustness analysis and the check of the closed forms. Notation as in the main text; all statistics come from `code/results/analysis/` (inputs of each output in `figtab/INPUTS.md`).

Table 4: Median tasks touched K [95% bootstrap interval] over the feasible targets, Open LLM Leaderboard (notation of Table 1, including "feas." and †).

rule	rank 6-10	rank 31-100	rank 101-500	rank >500
cap: `ceil`				
	(n=5)	(n=70)	(n=400)	(n=3996)
mean	1.0 [1.0, 1.0]	1.0 [1.0, 1.0]	1.0 [1.0, 1.0]	3.0 [2.0, 3.0]
geometric mean	1.0 [1.0, 1.0]	1.0 [1.0, 1.0]	2.0 [1.0, 2.0]	3.0 [3.0, 3.0]
trimmed mean	1.0 [1.0, 1.0]	1.0 [1.0, 1.0]	2.0 [2.0, 2.0]	3.0 [3.0, 3.0]
median	1.0 [0.0, 1.0]	2.0 [2.0, 2.0]	2.0 [2.0, 2.0]	3.0 [3.0, 3.0]
min	2.0 [2.0, 2.0]	2.0 [2.0, 2.0]	2.0 [2.0, 2.0]	3.0 [3.0, 3.0]
Borda*	2.0 [1.0, 2.0]	2.0 [2.0, 3.0]	4.0 [3.5, 4.0]	6.0 [6.0, 6.0]
Condorcet winner*	2.0 [2.0, 3.0]	3.0 [3.0, 4.0]	4.0 [4.0, 4.0]	4.0 [4.0, 4.0]
plurality*	2.0 [2.0, 2.0]	2.0 [2.0, 2.0]	2.0 [2.0, 2.0]	2.0 [2.0, 2.0]
cap: `emp95`				
	(n=5)	(n=70)	(n=400)	(n=3996)
mean	2.0 [2.0, 2.0]	4.0 [3.0, 4.0]	5.0 [4.0, 5.0] (236/400 feas.)	all inf. (3996)
geometric mean	4.0 [3.0, 6.0]	4.0 [3.0, 5.0] (28/70 feas.)	5.0 [5.0, 6.0] (60/400 feas.)	all inf. (3996)
trimmed mean	2.0 [1.0, 2.0]	3.0 [3.0, 4.0] (18/70 feas.)	4.0 [4.0, 4.0] (17/400 feas.)	all inf. (3996)
median	1.0 [0.0, 1.0]	2.0 [2.0, 3.0] (37/70 feas.)	3.0 [2.0, 3.0] (73/400 feas.)	4.0 [3.0, 4.0]† (4/3996 feas.)
min	2.0 [2.0, 2.0]† (1/5 feas.)	1.0 [1.0, 1.5] (14/70 feas.)	2.0 [1.0, 2.0] (25/400 feas.)	2.0 [2.0, 2.0] (15/3996 feas.)
Borda*	2.0 [2.0, 2.0]	2.0 [2.0, 3.0] (28/35 feas.)	4.0 [3.0, 4.5] (18/32 feas.)	all inf. (40)
Condorcet winner*	3.0 [3.0, 3.0]† (1/5 feas.)	5.0 [3.0, 5.0] (5/35 feas.)	all inf. (32)	all inf. (40)
plurality*	2.0 [2.0, 2.0]† (4/5 feas.)	2.0 [2.0, 2.0] (16/35 feas.)	2.0 [2.0, 2.0] (6/32 feas.)	all inf. (40)
cap: `gain10`				
	(n=5)	(n=70)	(n=400)	(n=3996)
mean	3.0 [3.0, 3.0]	6.0 [6.0, 6.0]	6.0 [6.0, 6.0] (14/400 feas.)	all inf. (3996)
geometric mean	3.0 [2.0, 3.0]	5.0 [4.0, 5.0] (53/70 feas.)	6.0 [6.0, 6.0] (60/400 feas.)	all inf. (3996)
trimmed mean	3.0 [3.0, 3.0]	4.0 [4.0, 4.0] (20/70 feas.)	5.0 [4.0, 5.0] (6/400 feas.)	all inf. (3996)
median	1.0 [0.0, 1.0]	2.5 [2.0, 3.0] (46/70 feas.)	3.0 [3.0, 3.0] (125/400 feas.)	4.0 [4.0, 4.0]† (3/3996 feas.)
min	2.0 [2.0, 2.0]† (4/5 feas.)	2.0 [2.0, 2.0] (46/70 feas.)	2.0 [2.0, 2.0] (280/400 feas.)	2.0 [2.0, 3.0] (98/3996 feas.)
Borda*	2.0 [1.0, 2.0]	3.0 [3.0, 3.0] (33/35 feas.)	4.0 [4.0, 5.0] (20/32 feas.)	all inf. (40)
Condorcet winner*	3.0 [3.0, 3.0]	4.5 [4.0, 5.0] (18/35 feas.)	6.0 [6.0, 6.0]† (1/32 feas.)	all inf. (40)
plurality*	all inf. (5)	2.0 [2.0, 2.0]† (2/35 feas.)	all inf. (32)	all inf. (40)

Table 5: Median total points L_1 [95% bootstrap interval] over the feasible targets, HELM MMLU v1.13.0 (79 models, 57 subjects; all rules on every non-leading target; notation of Table 1).

rule	rank 6-10	rank 11-30	rank 31-100
cap: `ceil`			
	(n=5)	(n=20)	(n=49)
mean	177 [156, 260]	435 [380, 481]	1021 [863, 1118]
geometric mean	164 [139, 239]	412 [357, 456]	992 [814, 1089]
trimmed mean	165 [155, 238]	408 [361, 469]	978 [818, 1081]
median	13 [0.4, 17]	38 [36, 56]	207 [169, 274]
min	24 [2.0, 26]	46 [27, 50]	200 [145, 248]
Borda	56 [54, 151]	286 [231, 319]	858 [678, 951]
Condorcet winner	29 [16, 62]	107 [100, 121]	312 [247, 364]
plurality	7.7 [3.5, 19]	25 [21, 27]	72 [58, 87]
cap: `gain10`			
	(n=5)	(n=20)	(n=49)
mean	177 [156, 260]	435 [380, 481]	all inf. (49)
geometric mean	174 [142, 257]	424 [339, 483] (15/20 feas.)	all inf. (49)
trimmed mean	165 [155, 238]	408 [361, 469]	all inf. (49)
median	13 [0.4, 17]	38 [36, 56]	87 [68, 103] (11/49 feas.)
min	2.2 [2.0, 2.4]† (2/5 feas.)	14 [8.3, 24] (6/20 feas.)	46 [42, 51]† (2/49 feas.)
Borda	56 [54, 151]	269 [203, 330] (16/20 feas.)	all inf. (49)
Condorcet winner	29 [16, 62]	107 [100, 121]	171 [159, 197] (9/49 feas.)
plurality	7.7 [3.5, 19]	25 [21, 27]	51 [46, 62] (28/49 feas.)

Table 6: Median tasks touched K [95% bootstrap interval] over the feasible targets, HELM MMLU v1.13.0 (notation of Table 1).

rule	rank 6-10	rank 11-30	rank 31-100
cap: `ceil`			
	(n=5)	(n=20)	(n=49)
mean	5.0 [4.0, 7.0]	10 [9.0, 12]	21 [18, 23]
geometric mean	4.0 [4.0, 7.0]	10 [9.0, 12]	21 [18, 23]
trimmed mean	5.0 [4.0, 7.0]	10 [9.0, 12]	21 [19, 23]
median	8.0 [1.0, 10]	14 [12, 15]	26 [23, 27]
min	2.0 [1.0, 4.0]	7.0 [5.5, 7.5]	17 [15, 18]
Borda	19 [15, 28]	38 [34, 41]	51 [49, 52]
Condorcet winner	18 [14, 23]	28 [28, 28]	29 [29, 29]
plurality	8.0 [5.0, 9.0]	9.0 [9.0, 9.0]	9.0 [9.0, 9.0]
cap: `gain10`			
	(n=5)	(n=20)	(n=49)
mean	18 [16, 27]	44 [39, 49]	all inf. (49)
geometric mean	18 [15, 26]	43 [35, 49] (15/20 feas.)	all inf. (49)
trimmed mean	17 [16, 24]	42 [37, 48]	all inf. (49)
median	8.0 [1.0, 10]	14 [12, 15]	19 [18, 21] (11/49 feas.)
min	1.5 [1.0, 2.0]† (2/5 feas.)	4.5 [2.5, 5.5] (6/20 feas.)	8.0 [8.0, 8.0]† (2/49 feas.)
Borda	21 [19, 33]	47 [40, 51] (16/20 feas.)	all inf. (49)
Condorcet winner	18 [14, 23]	28 [28, 28]	29 [28, 29] (9/49 feas.)
plurality	8.0 [5.0, 9.0]	9.0 [9.0, 9.0]	9.0 [9.0, 10] (28/49 feas.)

Table 7: Sensitivity to the strictness margin ε and to ties: median over the targets feasible in both runs of the ratio of L_1 at $\varepsilon = 0.1$ (left) and of the run counting ties as losses (right) to the main run ($\varepsilon = 10^{-3}$), with 95% bootstrap interval; "K chg." and "feas. chg." count the targets whose K or feasibility differs. *: the 136-target stratified sample.

		$\varepsilon = 0.1$ vs 10^{-3}			ties as loss		
data, cap	rule	L_1 ratio [CI]	K chg.	feas. chg.	L_1 ratio [CI]	K chg.	feas. chg.
OLLB, `gain10` (n=136)	Borda*	1.006 [1.005, 1.007]	0	0	1.000 [1.000, 1.000]	0	0
	plurality*	1.007 [1.006, 1.014]	0	0	1.000 [1.000, 1.000]	0	0
HELM, `ceil` (n=79)	Borda	1.010 [1.008, 1.014]	0	0	1.000 [1.000, 1.000]	0	0
	Condorcet winner	1.014 [1.011, 1.019]	0	0	1.000 [1.000, 1.000]	0	0
	mean	1.008 [1.007, 1.010]	10	0	–	–	–
	median	1.020 [1.014, 1.024]	8	0	–	–	–
	plurality	1.020 [1.017, 1.029]	0	0	1.000 [1.000, 1.000]	1	0
HELM, `gain10` (n=79)	Borda	1.029 [1.022, 1.047]	1	0	1.000 [1.000, 1.000]	6	1
	Condorcet winner	1.030 [1.025, 1.034]	0	0	1.000 [1.000, 1.000]	0	0
	mean	1.016 [1.013, 1.021]	14	1	–	–	–
	median	1.031 [1.027, 1.038]	6	0	–	–	–
	plurality	1.033 [1.026, 1.040]	4	0	1.000 [1.000, 1.000]	2	0

C Checks Added After Review

Composition of the empirical cap. Of the 540 pairs, 154 are typed chat (RLHF, DPO, IFT), 346 domain-specific fine-tunes, and 40 other (24 pretrained, 13 continuously pretrained, 2 merges, 1 other); the self-declared `Type` does not say whether a "domain" fine-tune is an instruction conversion, so we also drop pairs whose model name contains a chat-like token (263 pairs; 207 remain after removing both). The `emp95` values per task are, for all pairs / without chat-typed pairs / without chat-typed or chat-named pairs: IFEval 46.0 / 44.6 / 55.7, BBH 7.3 / 7.0 / 7.9, MATH Lvl 5 21.3 / 19.4 / 18.8, GPQA 3.8 / 3.9 / 3.1, MuSR 7.2 / 7.2 / 6.9, MMLU-PRO 3.2 / 3.2 / 3.2 (`cap_composition_pairs.csv`). Table 8 gives the share of targets that can become first. The ordering of the rules and the confinement of feasibility to roughly the top 500 targets hold under every variant; the absolute shares do not: across the definitions of Table 8 they range from 1.4 to 12.2% for the mean, 1.7 to 10.0% for the median, 0.4 to 16.0% for plurality and 0.8 to 6.1% for the minimum, which reaches 100% under `head0.25`. Not done: a cap calibrated on pairs verified from model cards as base-to-domain fine-tunes (no such data are supplied).

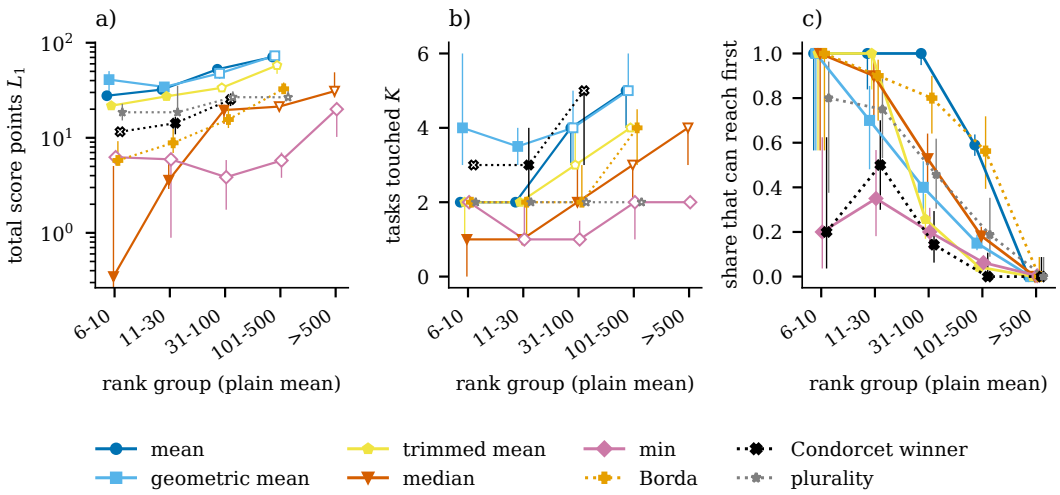


Figure 5: Investment at the empirical cap `emp95` on OLLB: (a) median L_1 and (b) median K over the targets of the rank group that can reach first (hollow marker: fewer than half of the group are feasible; no marker: none); (c) share of the group's targets that can (Wilson 95% interval). Groups: ranks 6–10, 11–30, 31–100, 101–500, >500 ($n = 5, 20, 70, 400, 3996$ for the score rules); ordinal rules (dotted) on the stratified sample. Bars in (a, b): 95% bootstrap intervals over targets.

Table 8: Share (%) of targets that can become first under cap variants on OLLB (plurality: stratified sample of 136 targets, stratum-weighted). `nochat`: emp95 without chat-typed pairs; `nochatname`: also without chat-named pairs; `frac95`: 95th percentile of the share of the remaining headroom closed by a fine-tune, applied to the target's own scores; `head0.25`: a quarter of the remaining headroom per task.

	`emp95`	`nochat`	`nochatname`	`emp99`	`frac95`	`head0.25`
mean	7.5	6.7	6.9	12.1	1.4	12.2
median	3.1	2.7	2.9	10.0	1.7	9.4
minimum	1.4	1.7	0.8	6.1	0.8	100
plurality	2.9	2.3	2.2	16.0	0.4	4.4

Fits for the growth with rank. For each rule and dataset we fitted the median L_1 at `ceil` against rank with a power law, a linear form and a logarithmic form, to the group medians and to the targets (`rank_dependence_fits.csv`). By the relative error (rmse of $\log_{10} L_1$) the power law is preferred in 25 of 32 OLLB and 16 of 18 HELM group-level fits, and in 21 of 32 OLLB and all 31 HELM target-level fits; by R^2 on the L_1 scale a linear form is preferred in 14 of 32 OLLB and 17 of 18 HELM group-level fits. For the mean on OLLB the exponent is 0.44 (R^2 0.954, against 0.980 for the linear form). The HELM fits rest on four or five groups and the exponent of 3.1 for the Condorcet requirement comes from near-zero costs at ranks 2–5. Group medians are monotone in rank in all 31 HELM cells and in 19 of 32 OLLB cells; for the median rule on OLLB the decrease (9.6 and 8.4 points at ranks 2 and 3–5, 0.3 at ranks 6–10) lies outside the bootstrap interval, for the minimum, harmonic mean, plurality and Condorcet requirement it lies within it. The first two groups hold one and three to five targets.

Duplicate model names. All 79 duplicated names are float16/bfloat16 pairs (36 share a submission date). Re-running the mean, median and minimum at `ceil` and the mean at `emp95` with the first, the latest or the worst-scoring variant instead of the best one leaves the leader and the runner-up unchanged for every rule and moves the group medians of L_1 by at most 5.3% (largest: minimum at ranks 31–100, 13.8 to 14.5 points; `latest` at most 2.2%); the share of infeasible targets at ranks 101–500 under the mean with `emp95` is 41% (best), 38% (first), 45% (latest) and 43% (worst) (`duplicate_rule_sensitivity.csv`, `duplicate_rule_leaders.csv`). The other robustness cells were not re-run.

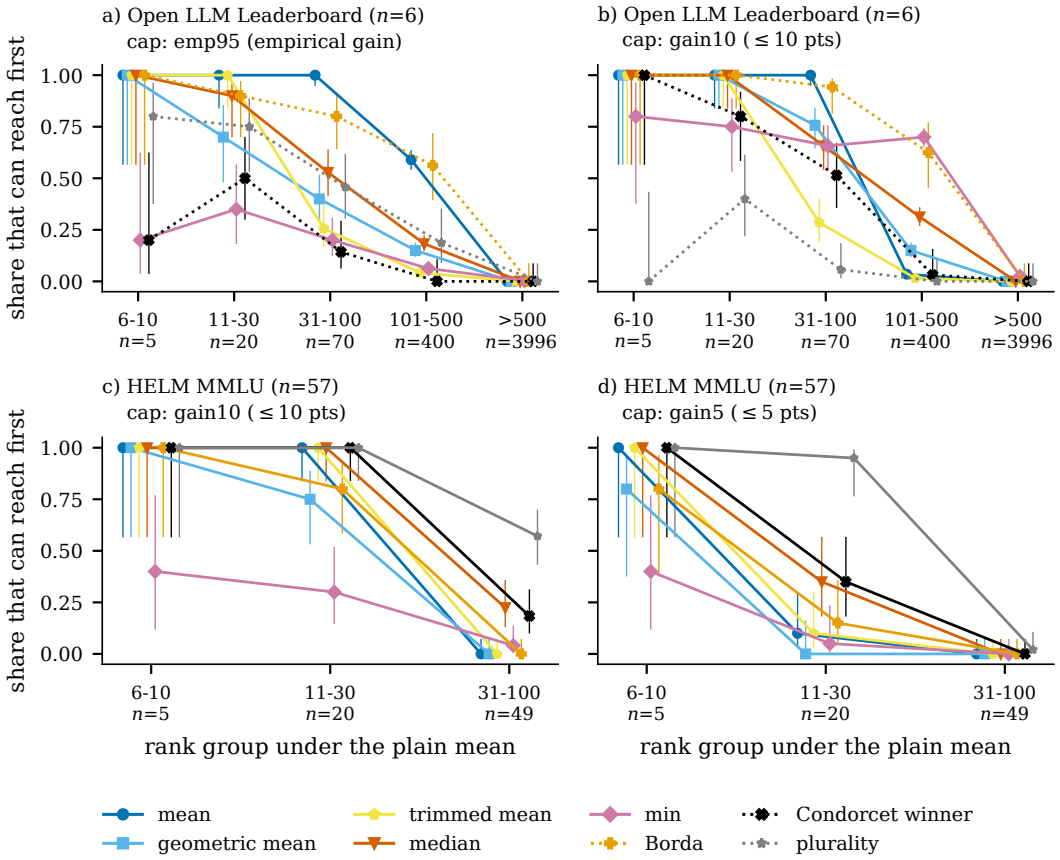


Figure 6: Share of targets in each rank group for which first place is reachable under a binding cap (bars: Wilson 95% intervals; infeasible targets are counted, not dropped). (a) OLLB, `emp95`; (b) OLLB, `gain10` (at most +10 points per task); (c, d) HELM, `gain10` and `gain5`. Under `ceil` and `front2` every target is feasible. Groups with fewer than five targets are omitted.

HELM test-set sizes. The 57 subjects have 100 to 1534 test items (10 subjects have 100, the median is 173, the total 14 042; identical for all 79 models), so one item is 1.0 point on a 100-item subject and 0.07 on the largest; the binomial standard error of a subject accuracy of 50% is 5.0, 3.8 and 1.3 points for 100, 173 and 1534 items (`helm_subject_sizes.csv`). A jump on one subject can thus be within evaluation noise for small subjects.

Score-cap defence. Clipping every model's per-task score at the leader's score plus h points ($h = 5, 10, 20$) leaves the share of targets that can become first at 100% for the mean, median and minimum on both datasets and the cost unchanged (median L_1 at $h = 5$: OLLB 180.5, 109.1, 43.2; HELM 696, 113, 98), because reaching the clip on every task is still enough. At $h = 0$ nobody can become strictly first, a tie rather than a higher price. Clips at the 99th, 95th or 90th percentile of each task also clip the leader; the share falls to 0 for the median and minimum on OLLB at all three levels, for the same rules on HELM at the 95th and 90th, and for the OLLB mean at the 90th (the HELM mean stays at 100%), but this is the strict-first rule meeting tied scores, not a higher cost of first place in points, so we do not use these rows for cost statements. Per-model clips and tie-allowed first place were not run (`score_cap_defence.csv`).

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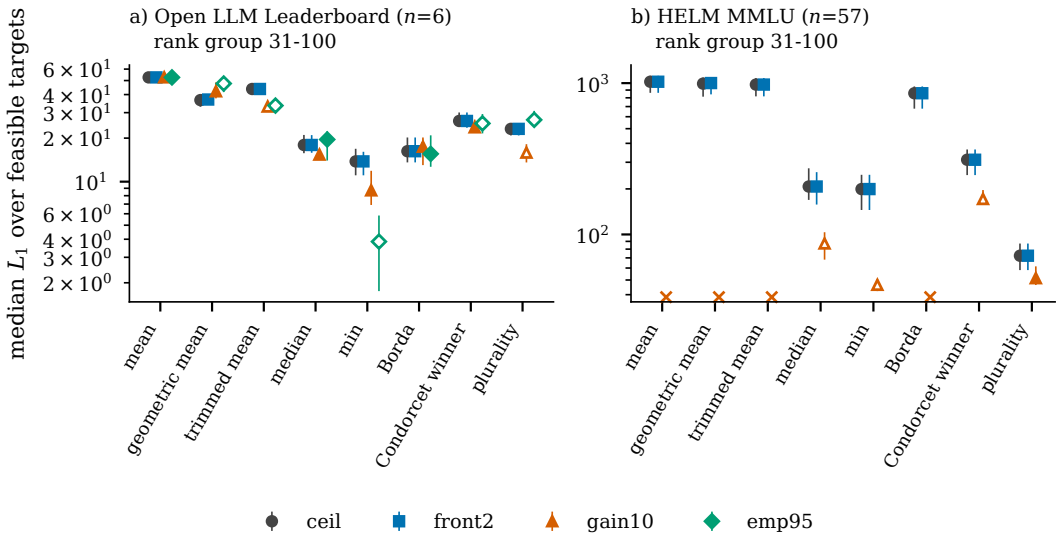


Figure 7: Median L_1 among the feasible targets of rank group 31–100 (OLLB $n = 70$, HELM $n = 49$) per rule and cap; bars: 95% bootstrap interval. Hollow marker: fewer than half of the group's targets are feasible under that cap (the median then describes the feasible minority only; see Figure 2 and the tables); $\times$ on the axis: no target is feasible. HELM has no `emp95`.

Answer: [Yes]

Justification: Abstract and Section 1 state the claims (exact minimum investment, rule dependence without a fixed order, targeted versus uniform cost, cap dependence) with their conditions (cap, single deviator, two leaderboards, ordinal rules on a stratified sample).

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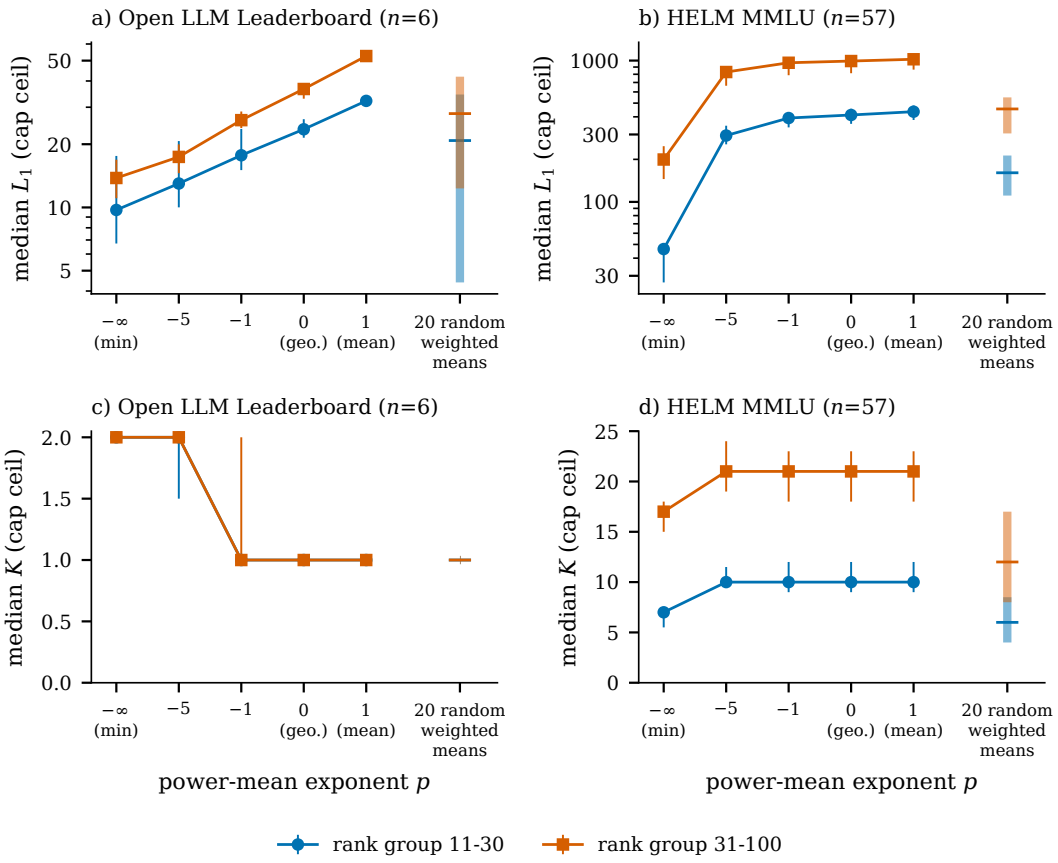


Figure 8: Median L_1 (a, b; log scale) and median K (c, d) along the power-mean continuum $p = -\infty$ (minimum), -5, -1 (harmonic), 0 (geometric), 1 (mean), cap `ceil`, rank groups 11–30 and 31–100 (95% bootstrap interval over targets); on OLLB K is 2 for $p \leq -5$ and 1 otherwise in both groups, so the two lines in panel c coincide. Right of each panel: spread over 20 random weighted means (Dirichlet weights, seeds $1000 + k$) of the group median; the bar spans the minimum to the maximum over the draws, the dash is their median.

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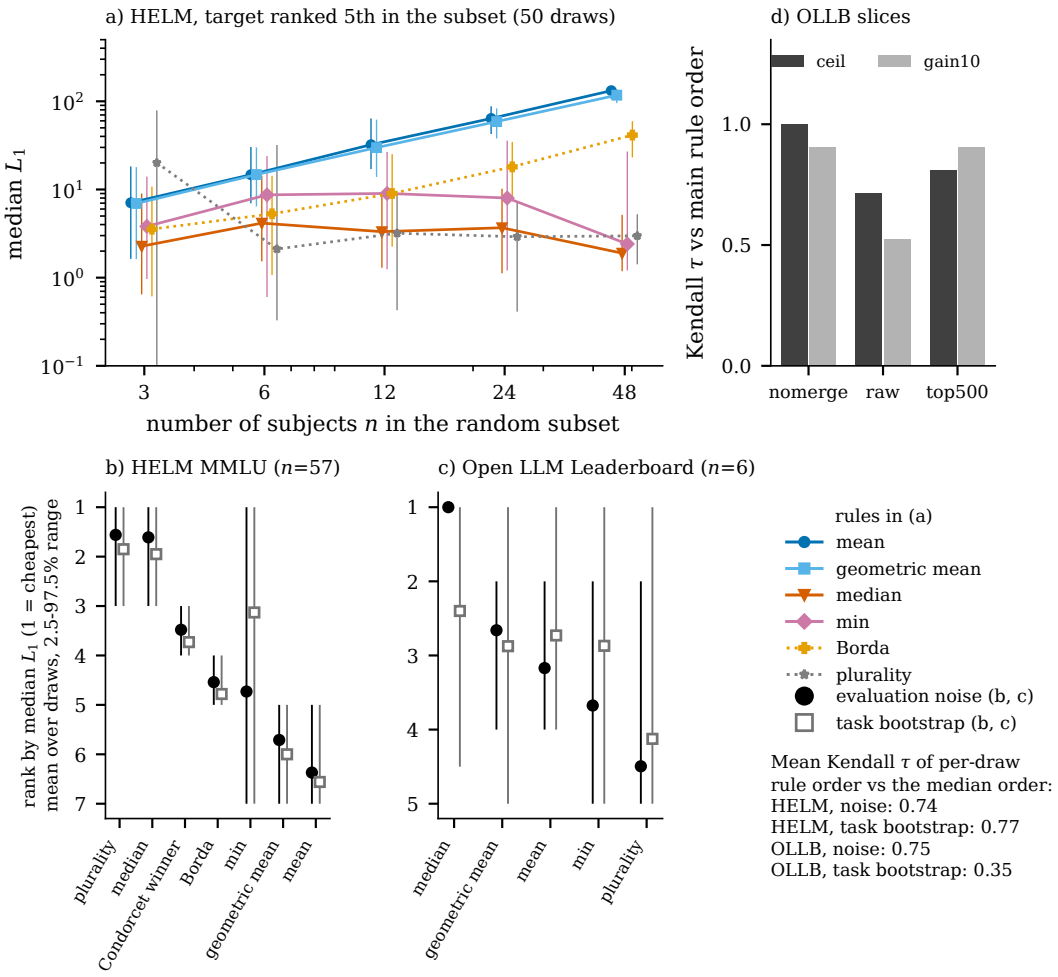


Figure 9: Robustness of the rule comparison. (a) HELM: median L_1 against the number n of subjects in a random subset (50 draws per n; target ranked fifth in the subset; bars: 2.5–97.5% range over draws; lower ends clipped at 0.1). (b, c) Rank of each rule by median L_1 among the ten best models after the leader, cap `gain10`, under evaluation noise and task bootstrap (100 draws each; mean rank with 2.5–97.5% range). (d) Kendall τ between the rule order on each OLLB data slice (no merging, raw instead of normalised scores, top 500 models; 60-target samples; `ceil` and `gain10`) and the main OLLB order.

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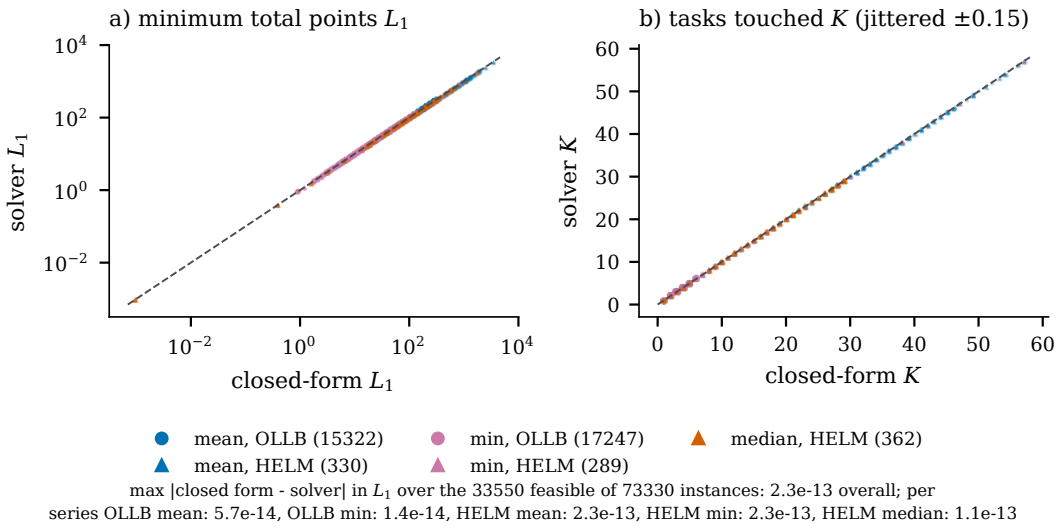


Figure 10: Closed-form prediction (coded independently of the solvers) against solver output for every instance (target × cap) of the mean, the median (HELM, odd n) and the minimum, on OLLB and HELM; dashed line: identity. (a) L_1 (log–log, feasible instances with positive prediction); (b) K (jittered). Infeasible instances are not plotted; feasibility agrees on all of them (`theory_checks.txt`).

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